

MITHILESH A

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PROFESSIONAL SUMMARY

High-achieving AI & Data Science student at SASTRA Deemed University specializing in **Reinforcement Learning (RL)**, **Deep Learning**, and **Quantitative Finance**. Quantitative Researcher at WorldQuant with a proven track record of achieving **95%+ accuracy** in financial forecasting models and developing intelligent infrastructure for the Smart India Hackathon. Adept at building scalable web solutions integrated with advanced neural architectures.

EDUCATION

SASTRA Deemed University

B.Tech in Computer Science (AI & Data Science)

Thanjavur, TN

Aug. 2024 – June 2028

- **Focus:** Reinforcement Learning, Multi-Agent Systems, and Algorithmic Trading.

Harvard University (CS50)

Web Programming with Python and JavaScript (CS50W)

Remote

In Progress

PROFESSIONAL & RESEARCH EXPERIENCE

WorldQuant

BRAIN Research Consultant

Remote

May 2026 – Present

- Developing quantitative alphas using WorldQuant's BRAIN platform and Custom Expression Languages.
- Applying probability theory and statistical modeling to construct and backtest financial signal strategies.
- Leveraging data science skills to research and validate alpha ideas across global financial markets.

WorldQuant

BRAIN Conditional Consultant

Remote

Apr. 2026 – May 2026

- Researched conditional alpha strategies using BRAIN's expression language and logic-based signal construction.
- Worked with probability-driven frameworks to evaluate signal quality and market applicability.

Student Researcher (Financial Markets)

DNN-RL Based Solution for Predicting Exchange Rates

SASTRA Deemed University

Mar. 2026 – Present

- **Submitted a specialized quantitative research paper to IEEE; currently awaiting peer review.**
- Integrating Deep Learning and policy-gradient Reinforcement Learning (PPO) to augment time series models.
- Incorporated critical macroeconomic variables including Crude Oil prices, Gold futures, and the NIFTY 50 index to achieve a predictive accuracy exceeding **95%**.

Student Researcher (Cybersecurity)

Advanced Persistent Threat (APT) Detection

SASTRA Deemed University

Feb. 2026 – Present

- Investigating APT detection methodologies by leveraging deep learning (DL) models to analyze network traffic and identify distinct stages of cyber vulnerabilities.

TECHNICAL PROJECTS

RL-Based Traffic Management (SIH 2025) | DQN, MARL, OpenCV, Raspberry Pi

Sep. 2025 – Dec. 2025

- Developed an intelligent urban signal control system for the Smart India Hackathon (Odisha) to minimize traffic clearance time.
- Integrated OpenCV on Raspberry Pi for real-time vehicle detection, preprocessing, and signal adjustments.

Web-Based AI Applications | Python, Django, MongoDB, ML

Jan. 2026 – Present

- Engineered various full-stack web platforms integrating custom AI/ML-backed solutions for seamless user interaction.

TECHNICAL SKILLS

Programming: Python (Advanced), Java (Swing/GUI), C++, C, SQL, JavaScript

AI/ML/Data: Deep Learning, Machine Learning, Reinforcement Learning (PPO, DQN, MARL), TensorFlow, OpenCV, NumPy, Pandas, Matplotlib, Seaborn

Web/Tools: Python Stack (Django), HTML/CSS, Git, MongoDB, Feature Engineering, Data Visualization

Domain Knowledge: High Performance Computing (HPC), Financial Markets, Quantitative Finance, Macroeconomic Analysis, Cybersecurity, Data Structures & Algorithms (DSA)

CERTIFICATIONS & ACHIEVEMENTS

- **NPTEL SWAYAM:** Completed **Deep Learning Specialization**, mastering neural network architectures.
- **CMI STEMS:** Selected in the **Top 30 in India** for research aptitude by Chennai Mathematical Institute (2025).
- **Yale University:** Completed *Financial Markets* certification via Coursera, focusing on market theory.
- **Mathematics Olympiad:** Merit certificate holder in the Indian Olympiad Qualifier in Mathematics (IOQM).